

Problem Set 2 - Responses

Tate Mason

Paper 1 - “Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records” - Joshua D. Angrist (1990)

Question 1: What is the research Question

How did being in the Vietnam war affect labor outcomes, specifically earnings? They induce randomization via the draft lottery, implying there was no endogeneity since people were selected, ostensibly, at random.

Question 2: Why is it important to answer such a question?

This is an important question for both moral and political reasons. Morally, it is important to know if soldiers are fairly compensated after their service. There should not be a penalty for service, and thus understanding how being a soldier impacts labor outcomes is important. Policy-wise, understanding labor market outcomes is important when implementing legislation which is targeted at soldiers and veterans.

Question 3: What are the key endogeneity concerns?

Endogeneity concerns lie in health status as well as education status. For instance, healthy individuals are likely to be eligible, while those in college could likely defer service. Thus, units being healthier or more educated, two covariates which impact earning potential, could be endogenous to the treatment.

Question 4: What is the argument for the relevance of the IV of the paper? Does the paper show any evidence in favor of that?

The paper argues that the draft lottery is relevant because it is correlated with serving in Vietnam.

Question 5: What is the argument for the validity of the IV of the paper? Does the paper show any evidence in favor of that?

The paper argues that the draft lottery is valid because it is random, and thus uncorrelated with earnings directly, only through the channel of serving in Vietnam.

Question 6: What is the argument for the monotonicity of the IV of the paper? Does the paper show any evidence in favor of that?

There are no defiers here, as while one can defy the draft, one cannot defy the lottery. The assumption is reasonable given the nature of the instrument and thus does not require argumentation.

Question 7: The IV approach estimates the LATE not the ATE. Can you circle back to items 1 and 2 and see if your answer is still the same for LATE, not for ATE?

Yes, the answers to questions 1 and 2 still hold for LATE. The research question is still important because it is important to understand how serving in Vietnam impacts earnings for those who were actually affected by the draft lottery. The importance of the question also remains because understanding the labor market outcomes for those who served in Vietnam is crucial for both moral and policy reasons.

Question 8: Can you criticize the validity and the monotonicity assumptions? What are the most realistic violations of these two assumptions you can think of?

Monotonicity is quickly violated by draft dodgers and those who enlist, a group they discuss themselves. Validity is more difficult to violate here, due to the random nature of being drafted.

Question 9: Related to question 8, can you think of additional evidence the paper could have shown to check whether your criticisms are important for the conclusions of the paper?

The paper could have provided more evidence in favor of the validity of the instrument by showing that the draft lottery is uncorrelated with other covariates that may impact earnings. For example, they could have shown that there are no significant differences in health status or education status across different draft lottery numbers. Further, there could have been more discussion on the

impact of draft dodgers and those who enlist, and how these groups may impact the results.

Question 10: What is the strongest part of the paper in your view?

The strongest part of the paper is the use of the draft lottery as an instrument. The draft lottery is a well-known natural experiment, and its random nature makes it a strong candidate for an instrument. Additionally, the paper does a good job of addressing potential endogeneity concerns and providing evidence in favor of the relevance and validity of the instrument.

Question 11: What is the weakest part of the paper in your view?

The weakest part of the paper is the lack of discussion on potential violations of the monotonicity assumption. While the paper does mention draft dodgers and those who enlist, it does not provide any evidence or discussion on how these groups may impact the results. Additionally, the paper could have provided more evidence in favor of the validity of the instrument, such as showing that the draft lottery is uncorrelated with other covariates that may impact earnings.

Paper 2 - “Does Compulsory School Attendance Affect Schooling and Earnings?” - Joshua D. Angrist and Alan B. Krueger (1991)

Question 1: What is the Research Question

The authors seek to understand the impact of compulsory schooling policy on educational and earnings outcomes. They use quarter of birth as an instrument, arguing that those born in different quarters are subject to different compulsory schooling laws, which in turn impact their educational attainment and earnings.

Question 2: Why is it important to answer such a question?

This is an important question because it informs policy decisions regarding compulsory schooling laws, and is important for understanding early human capital formation. Understanding the impact of compulsory schooling on educational and earnings outcomes is crucial for policymakers to make informed decisions about education policy.

Question 3: What are the key endogeneity concerns?

The only one that comes to mind is that quarter of birth could be correlated with other factors that impact educational and earnings outcomes, such as seasonality

in health. However, the authors argue that quarter of birth is random and thus uncorrelated with these factors, which I agree with. It is a convincing argument.

Question 4: What is the argument for the relevance of the IV of the paper? Does the paper show any evidence in favor of that?

As stated above, the authors argue that quarter of birth is relevant because it is correlated with compulsory schooling laws, which in turn impact educational attainment and earnings. The paper provides evidence in favor of this argument by showing that those born in different quarters have different educational attainment and earnings outcomes. Further, they show that those born in the first quarter of the year are more likely to drop out of school earlier than those born in later quarters, which is consistent with the idea that compulsory schooling laws impact educational attainment.

Question 5: What is the argument for the validity of the IV of the paper? Does the paper show any evidence in favor of that?

The authors argue that quarter of birth is valid because it is random and thus uncorrelated with other factors that may impact educational and earnings outcomes. The paper provides evidence in favor of this argument by showing that there are no significant differences in observable characteristics, such as family income and parental education, across different quarters of birth. This suggests that quarter of birth is not correlated with other factors that may impact educational and earnings outcomes.

Question 6: What is the argument for the monotonicity of the IV of the paper? Does the paper show any evidence in favor of that?

The authors argue that quarter of birth is monotonic because there are no defiers. The assumption is reasonable given the nature of the instrument and thus does not require argumentation.

Question 7: The IV approach estimates the LATE not the ATE. Can you circle back to items 1 and 2 and see if your answer is still the same for LATE, not for ATE?

Yes, the answers to questions 1 and 2 still hold for LATE. The research question is still important because it is important to understand how compulsory schooling laws impact educational and earnings outcomes for those who are actually affected by the laws. The importance of the question also remains because understanding the impact of compulsory schooling on educational and earnings

outcomes is crucial for policymakers to make informed decisions about education policy.

Question 8: Can you criticize the validity and the monotonicity assumptions? What are the most realistic violations of these two assumptions you can think of?

Monotonicity is difficult to violate here, as there are no defiers. Validity could be violated if quarter of birth is correlated with other factors that impact educational and earnings outcomes, like health seasonality. However, the authors provide evidence in favor of the validity of the instrument by showing that there are no significant differences in observable characteristics across different quarters of birth.

Question 9: Related to question 8, can you think of additional evidence the paper could have shown to check whether your criticisms are important for the conclusions of the paper?

The paper could have provided more evidence in favor of the validity of the instrument by showing that quarter of birth is uncorrelated with other covariates that may impact educational and earnings outcomes. For example, they could have shown that there are no significant differences in health outcomes across different quarters of birth.

Question 10: What is the strongest part of the paper in your view?

The strongest part of the paper is the instrument itself. Quarter of birth is a well-known natural experiment, and its random nature makes it a strong candidate for an instrument. Additionally, the paper does a good job of addressing potential endogeneity concerns and providing evidence in favor of the relevance and validity of the instrument.

Question 11: What is the weakest part of the paper in your view?

The weakest part of the paper is the lack of discussion on potential violations of the monotonicity assumption. While the authors argue that there are no defiers, they do not provide any evidence or discussion on how this assumption may impact the results.

Paper 3 - “Children and Their Parents’ Labor Supply: Evidence from Exogenous Variation in Family Size” - Joshua D. Angrist and William N. Evans (1998)

Question 1: What is the Research Question

The authors seek to understand the impact of family size on parental labor supply. They use sex composition of the first two children as instruments, arguing that these factors are random and thus uncorrelated with other factors that may impact parental labor supply.

Question 2: Why is it important to answer such a question?

This is an important question for labor economics and family economics. Understanding the impact of family size on parental labor supply is crucial for policymakers to make informed decisions about family policy, such as childcare subsidies and parental leave policies, especially for poorer families. Additionally, understanding the impact of family size on parental labor supply is important for understanding the trade-offs that parents face when making decisions about work and family.

Question 3: What are the key endogeneity concerns?

The key endogeneity concern is that family size may be correlated with labor preferences which in turn impact labor supply. For example, parents who prefer to work less may choose to have more children.

Question 4: What is the argument for the relevance of the IV of the paper? Does the paper show any evidence in favor of that?

The authors argue that twin births and sex composition of the first two children are relevant because they are correlated with family size, which in turn impacts parental labor supply. The paper provides evidence in favor of this argument by showing that families with twins or with two children of the same sex are more likely to have more children than families without these characteristics.

Question 5: What is the argument for the validity of the IV of the paper? Does the paper show any evidence in favor of that?

The authors argue that sex composition of the first two children are valid because they are random and thus uncorrelated with other factors that may

impact parental labor supply. The paper provides evidence in favor of this argument by showing that there are no significant differences in observable characteristics, such as parental education and income, across families with and without twins or with different sex compositions.

Question 6: What is the argument for the monotonicity of the IV of the paper? Does the paper show any evidence in favor of that?

The authors argue that sex composition of the first two children are monotonic because there is no way to defy the instrument. The assumption is reasonable given the nature of the instrument and thus does not require argumentation.

Question 7: The IV approach estimates the LATE not the ATE. Can you circle back to items 1 and 2 and see if your answer is still the same for LATE, not for ATE?

Yes, the answers to questions 1 and 2 still hold for LATE. The research question is still important because it is important to understand how family size impacts parental labor supply for those who are actually affected by the instruments. The importance of the question also remains because understanding the impact of family size on parental labor supply is crucial for policymakers to make informed decisions about family policy.

Question 8: Can you criticize the validity and the monotonicity assumptions? What are the most realistic violations of these two assumptions you can think of?

Monotonicity could only be violated if there were some way to defy the instrument, which is not possible here. Validity could be violated if sex composition are correlated with other factors that impact parental labor supply, such as parental preferences for work. However, the authors provide evidence in favor of the validity of the instruments by showing that there are no significant differences in observable characteristics across families with and without twins or with different sex compositions.

Question 9: Related to question 8, can you think of additional evidence the paper could have shown to check whether your criticisms are important for the conclusions of the paper?

The paper actually handled these criticisms in the paper, showing that observable characteristics are similar across groups. They could have also shown that there are no significant differences in unobservable characteristics, such as parental preferences for work, across families with different sex compositions. The main

issue is the strength of the instrument, which they do discuss as not being able to explain everything about labor outcomes.

Question 10: What is the strongest part of the paper in your view?

The strongest part of the paper is the instrument, the data, and their strategy. The use of sex composition of the first two children as an instrument is clever and well-suited to the research question. Additionally, the paper does a good job of addressing potential endogeneity concerns and providing evidence in favor of the relevance and validity of the instruments. Further, the data is rich and allows for a thorough analysis of the research question.

Question 11: What is the weakest part of the paper in your view?

The weakness of the paper is the weakness of the instrument. While the instrument is clever and well-suited to the research question, it is not a very strong instrument. The authors acknowledge this limitation and discuss it in the paper, but it is still a weakness of the study. Additionally, the paper could have provided more evidence in favor of the validity of the instruments, such as showing that there are no significant differences in unobservable characteristics across families with different sex compositions.

**Paper 4 - “Using Geographic Variation in College Proximity to Estimate the Return to Schooling”
- David Card (1993)**

Question 1: What is the Research Question

The author seeks to understand the impact of college proximity on educational attainment and earnings. He uses geographic variation in college proximity as an instrument, arguing that those who live closer to colleges are more likely to attend college, which in turn impacts their educational attainment and earnings.

Question 2: Why is it important to answer such a question?

This is an important question because it informs policy decisions regarding college access and affordability. Understanding the impact of college proximity on educational attainment and earnings is crucial for policymakers to make informed decisions about education policy, such as funding for public colleges and universities.

Question 3: What are the key endogeneity concerns?

Key endogeneity concerns include college proximity being correlated with familial factors like parental education, which could encourage both living near a college and higher educational attainment. Another concern is that college proximity could be correlated with local labor market conditions, which could impact earnings. However, the author controls for this and makes the argument with mainly the lowest income quartile, which should alleviate this concern. Further, there is an issue of educational quality in primary schooling, which could be correlated with college proximity and impact educational attainment and earnings.

Question 4: What is the argument for the relevance of the IV of the paper? Does the paper show any evidence in favor of that?

The author argues that college proximity is relevant because it is correlated with college attendance, which in turn impacts educational attainment and through that channel, earnings. The paper provides evidence in favor of this argument by showing that proximity to a college is positively correlated with earnings outcomes, especially for those in the lowest income quartile.

Question 5: What is the argument for the validity of the IV of the paper? Does the paper show any evidence in favor of that?

The author argues that college proximity is valid because it is random and thus uncorrelated with other factors that may impact educational attainment and earnings. The paper acknowledges that it relies on restrictive assumptions for validity, but argues that controlling for local labor market conditions and focusing on the lowest income quartile helps to alleviate concerns about correlation with other factors.

Question 6: What is the argument for the monotonicity of the IV of the paper? Does the paper show any evidence in favor of that?

College proximity is monotonic because defying the instrument would not be observable. If someone lived near a college and chose not to go, we would not know if they were a defier or just had other reasons for not attending. The assumption is reasonable given the nature of the instrument.

Question 7: The IV approach estimates the LATE not the ATE. Can you circle back to items 1 and 2 and see if your answer is still the same for LATE, not for ATE?

Yes, the answers to questions 1 and 2 still hold for LATE. The research question is still important because it is important to understand how college proximity impacts educational attainment and earnings for those who are actually affected by the instrument. The importance of the question also remains because understanding the impact of college proximity on educational attainment and earnings is crucial for policymakers to make informed decisions about education policy.

Question 8: Can you criticize the validity and the monotonicity assumptions? What are the most realistic violations of these two assumptions you can think of?

Monotonicity is violated if there are defiers, which is possible but not observable. Validity could be violated if college proximity is correlated with other factors that impact educational attainment and earnings, such as familial factors like parental education or local labor market conditions. However, the author makes assumptions which help alleviate these concerns. The argument is imperfect, but reasonable.

Question 9: Related to question 8, can you think of additional evidence the paper could have shown to check whether your criticisms are important for the conclusions of the paper?

Given the time of the paper, I actually think its arguments were strong. However, through a modern lens, the paper could have provided more controls for familial factors. Further, the paper could have provided more evidence in favor of the validity of the instrument by showing that college proximity is uncorrelated with other covariates that may impact educational attainment and earnings.

Question 10: What is the strongest part of the paper in your view?

The strongest part of the paper is its instrument. College proximity is a clever and well-suited instrument for the research question. Additionally, the paper does a good job of addressing potential endogeneity concerns and providing evidence in favor of the relevance of the instrument.

Question 11: What is the weakest part of the paper in your view?

The weakest part of the paper is the assumptions needed for estimation. While the author makes reasonable arguments for the validity of the instrument, the assumptions are still restrictive and may not hold in all cases. Additionally, the paper could have provided more evidence in favor of the validity of the instrument, such as showing that there are no significant differences in unobservable characteristics across individuals with different levels of college proximity.